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Yang et al.

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(54) **COVER DEVICE**

(71) Applicant: **Universal Trim Supply Co., Ltd.**, New Taipei (TW)

(72) Inventors: **Shih-Sheng Yang**, Taipei (TW); **Chih-Wei Yang**, Taipei (TW)

(73) Assignee: **Universal Trim Supply Co., Ltd.**, New Taipei (TW)

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B65D 43/02 (2006.01)
B65D 81/20 (2006.01)

(52) **U.S. Cl.**
CPC ***B65D 51/24*** (2013.01); ***B65D 43/0202*** (2013.01); ***B65D 81/2007*** (2013.01); *B65D 2543/00092* (2013.01)

(58) **Field of Classification Search**
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See application file for complete search history.

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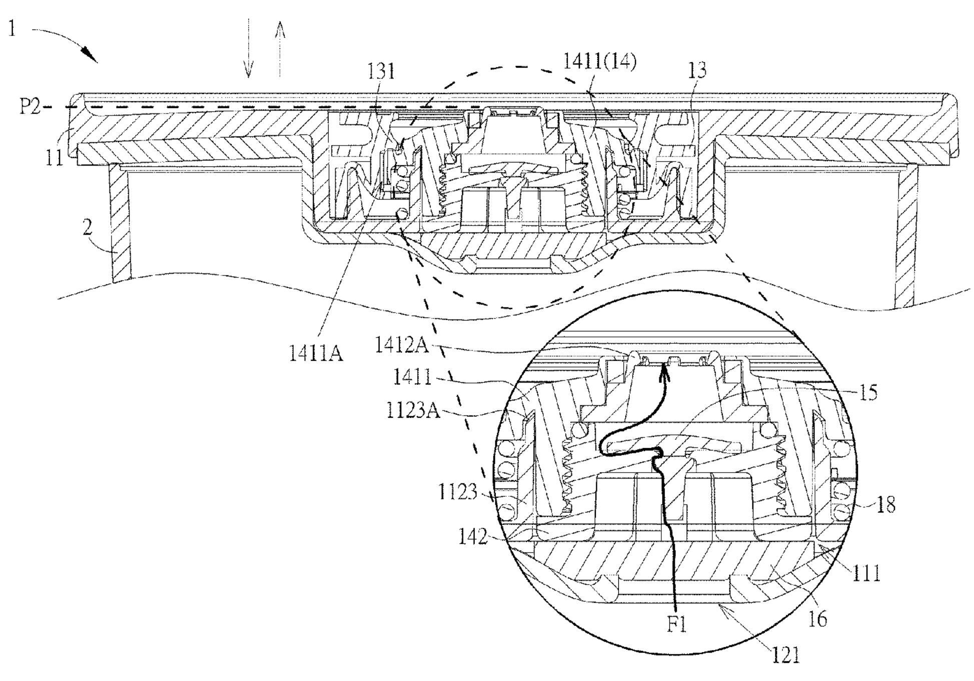
Primary Examiner — Jessica Cahill
(74) *Attorney, Agent, or Firm* — Winston Hsu

(57) **ABSTRACT**

A cover device is provided and includes a first body, a second body, a resilient component, a seat assembly, a check valve and a filtering component. The second body is removably attached on the first body and resiliently deformable. The resilient component is disposed on the first body. The seat assembly is movable relative to the first body. The seat assembly includes a first seat and a second seat. The first seat is removably engaged with the resilient component and configured to be connected to a vacuum device. The second seat is removably engaged with the first seat. The check valve is disposed on the second seat. The filtering component is disposed between the second seat and the second body.

20 Claims, 9 Drawing Sheets



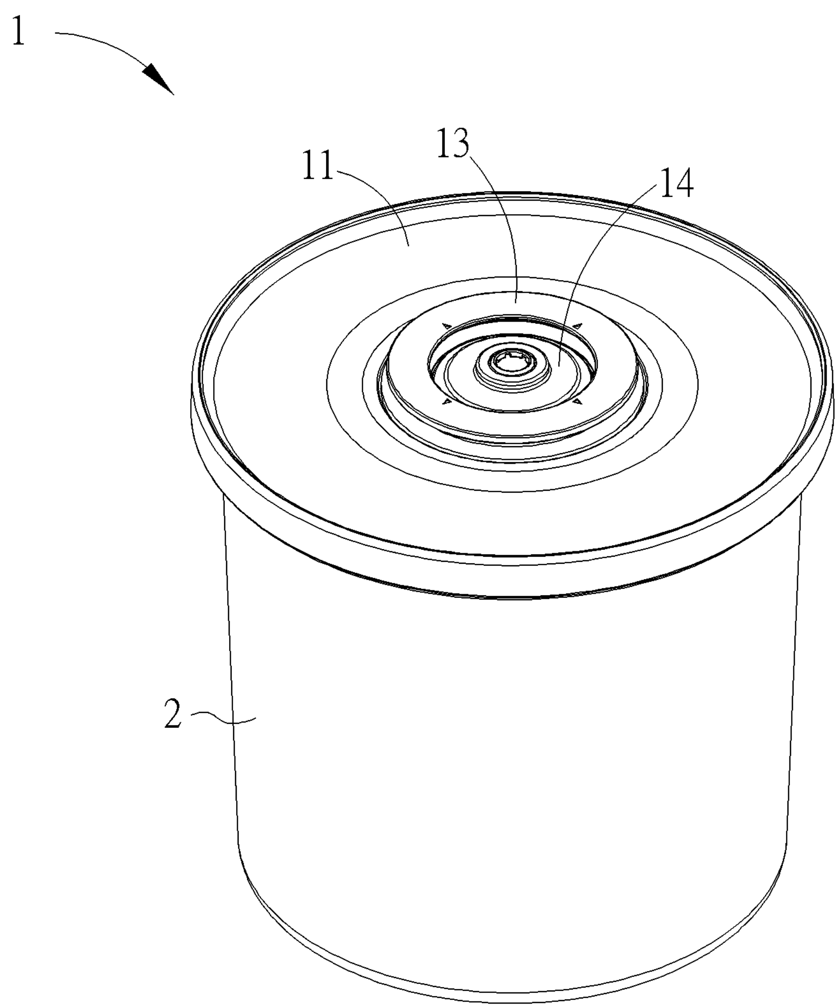


FIG. 1

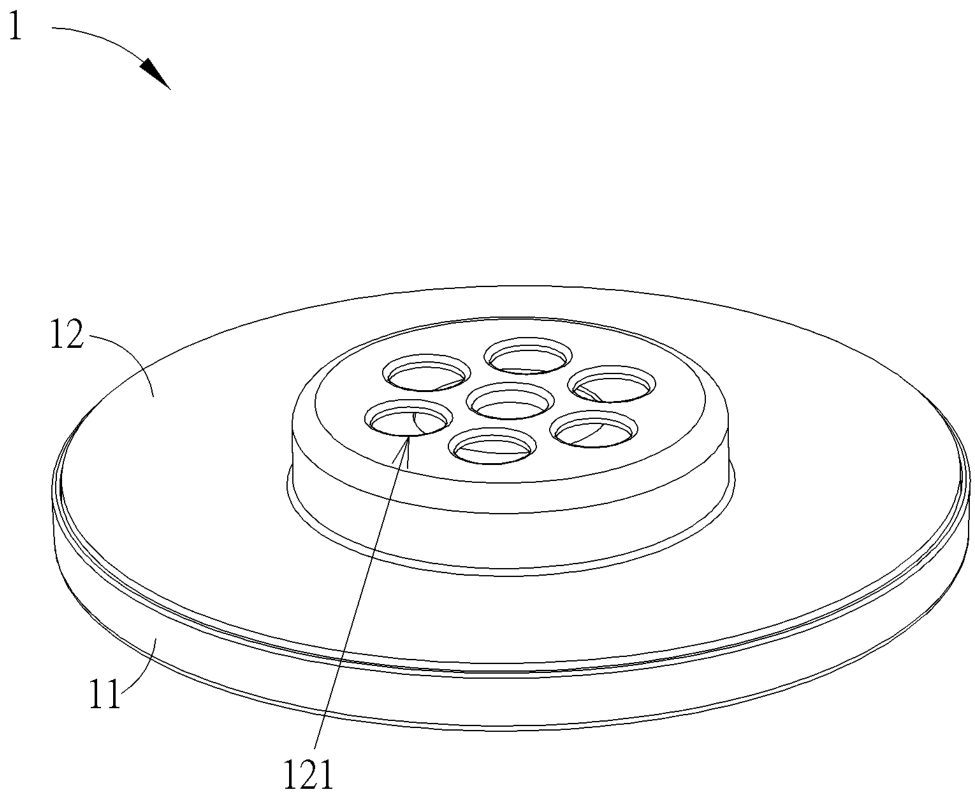


FIG. 2

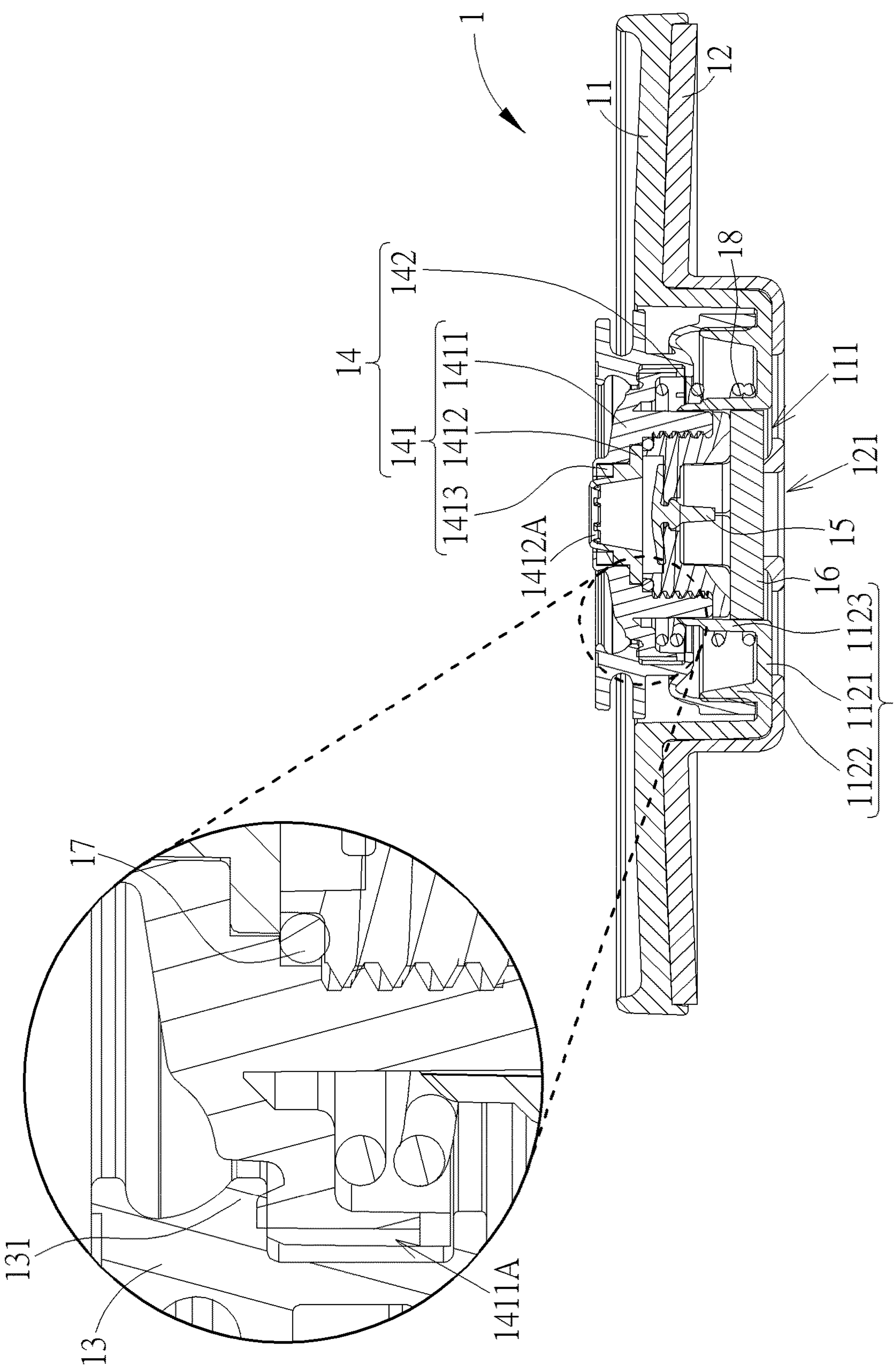

1
11
12
14
141
142
1411
1412
1413
1412A
18
111
121
15
16
112
1121
1122
1123
17
131
13
1411A

FIG. 3

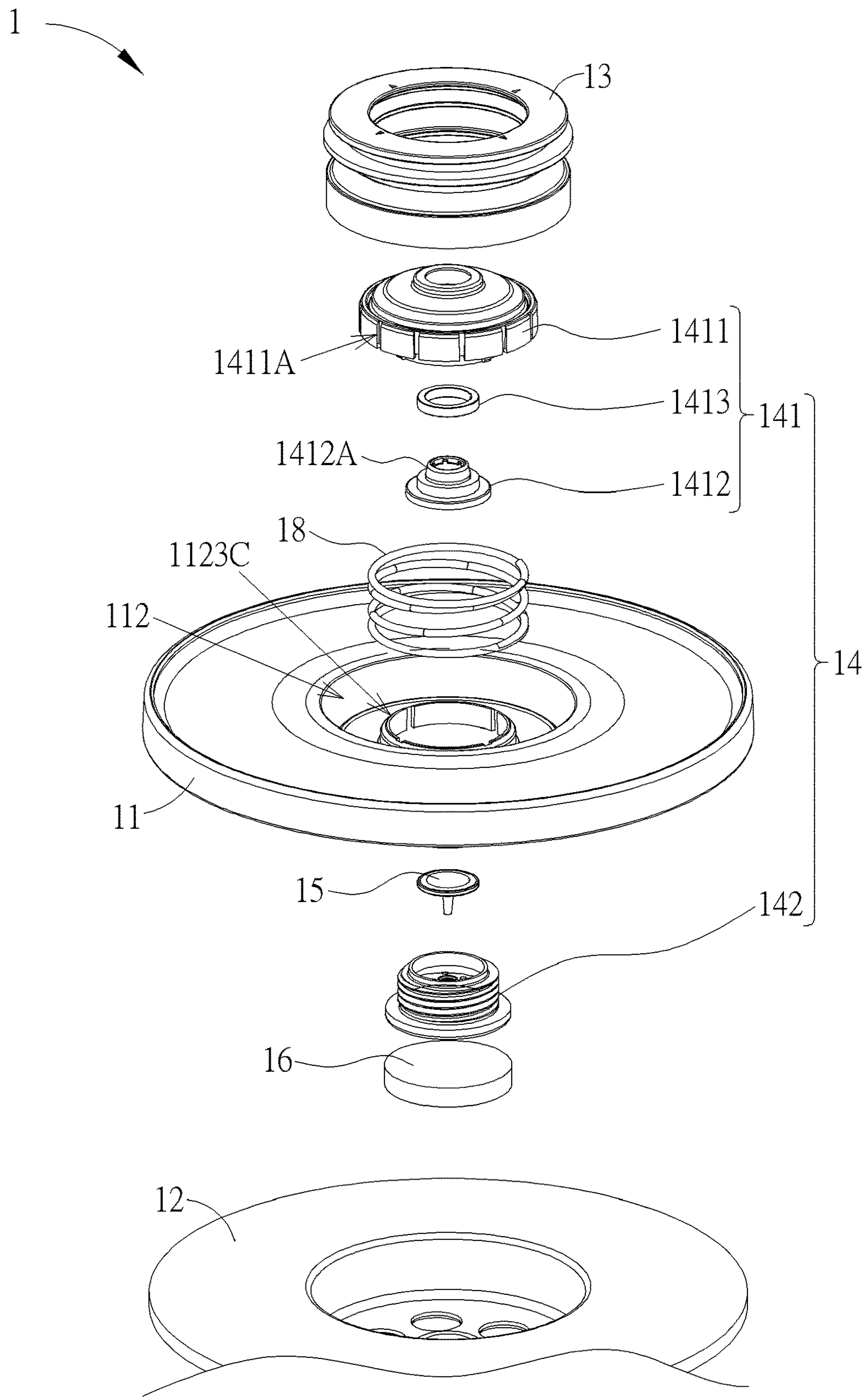

1
13
1411
1411A
1413
141
1412A
1412
18
1123C
112
14
11
15
142
16
12

FIG. 4

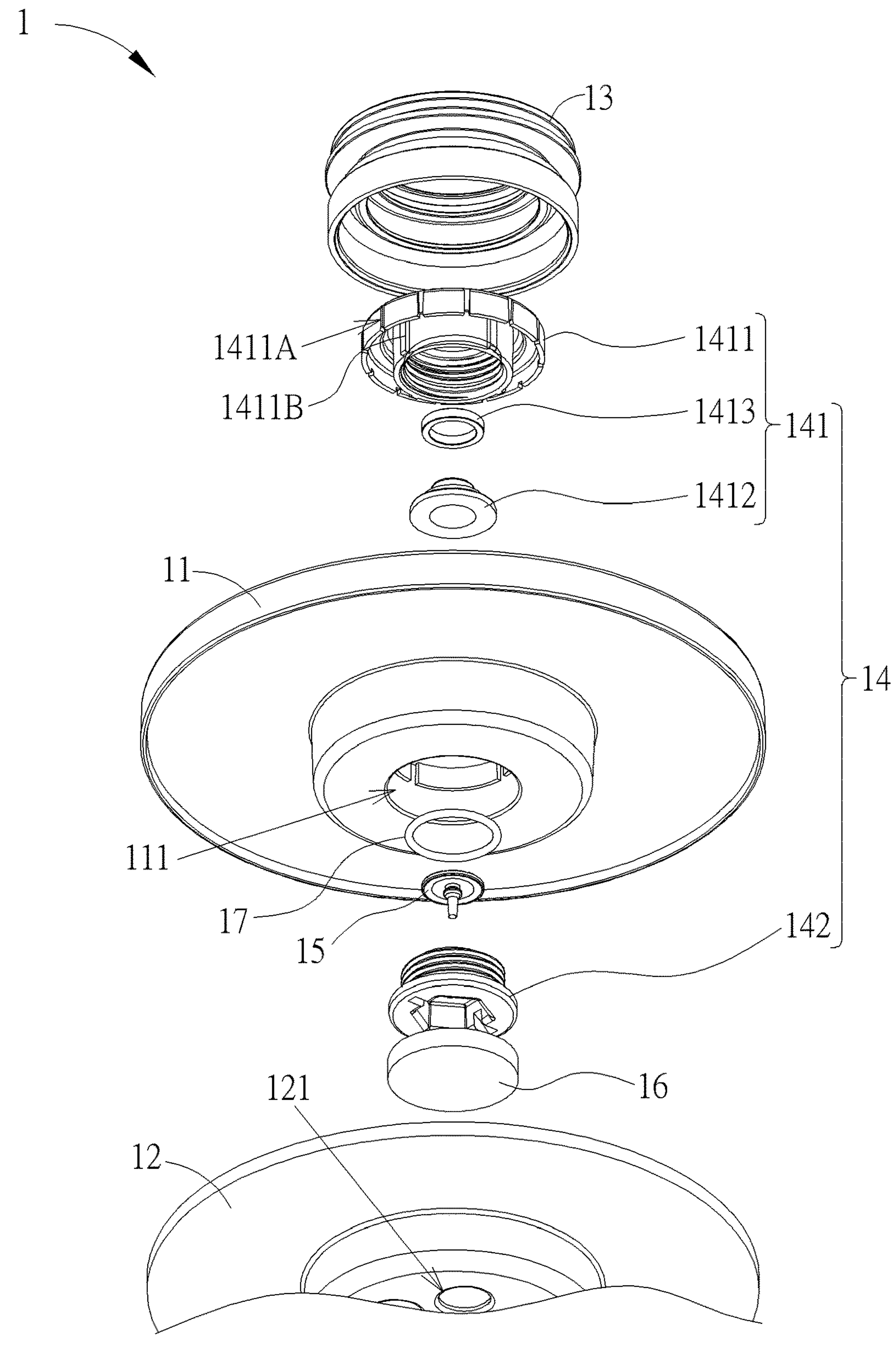


FIG. 5

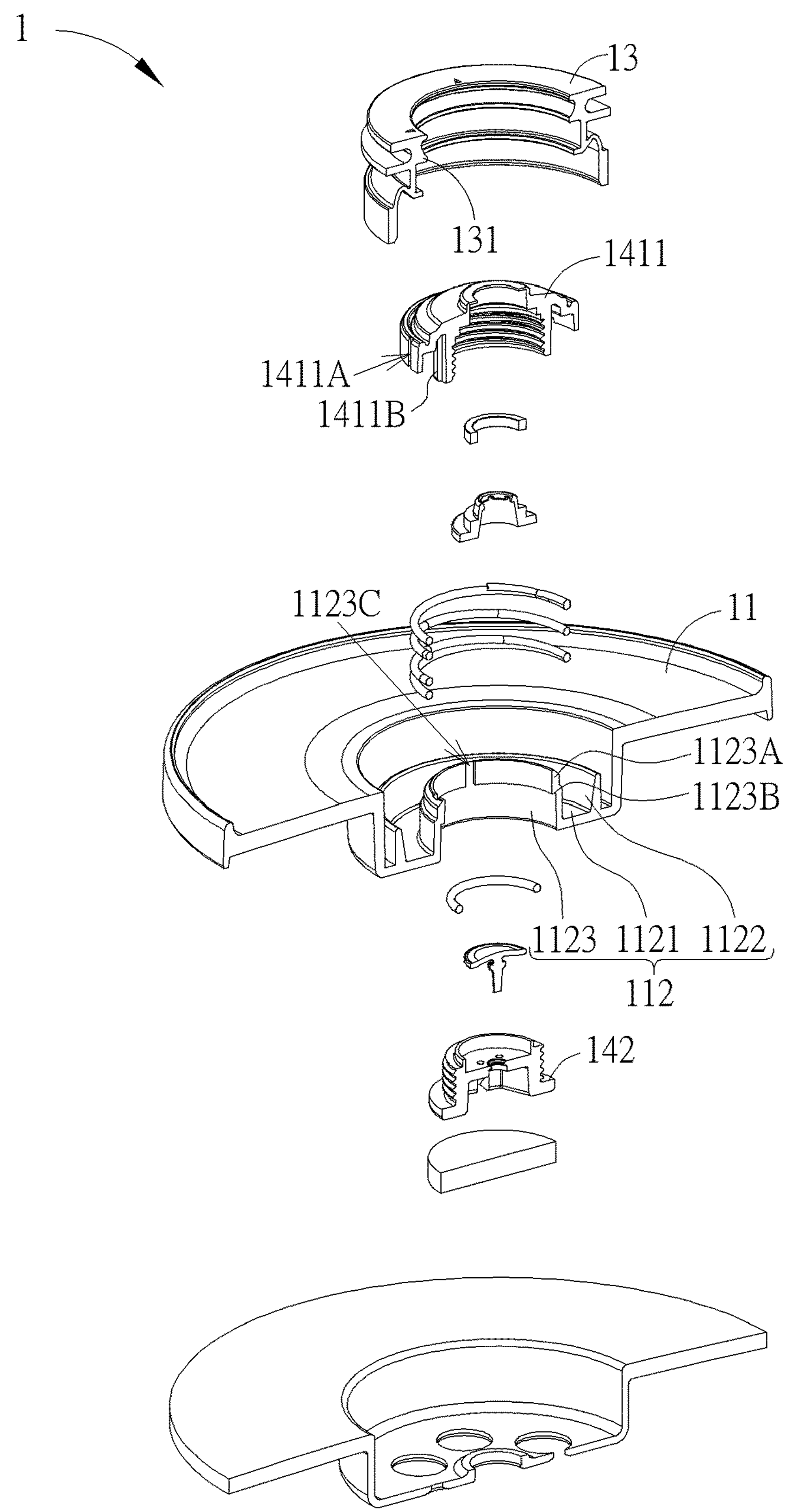

1
13
131
1411
1411A
1411B
1123C
11
1123A
1123B
1123
1121
1122
112
142

FIG. 6

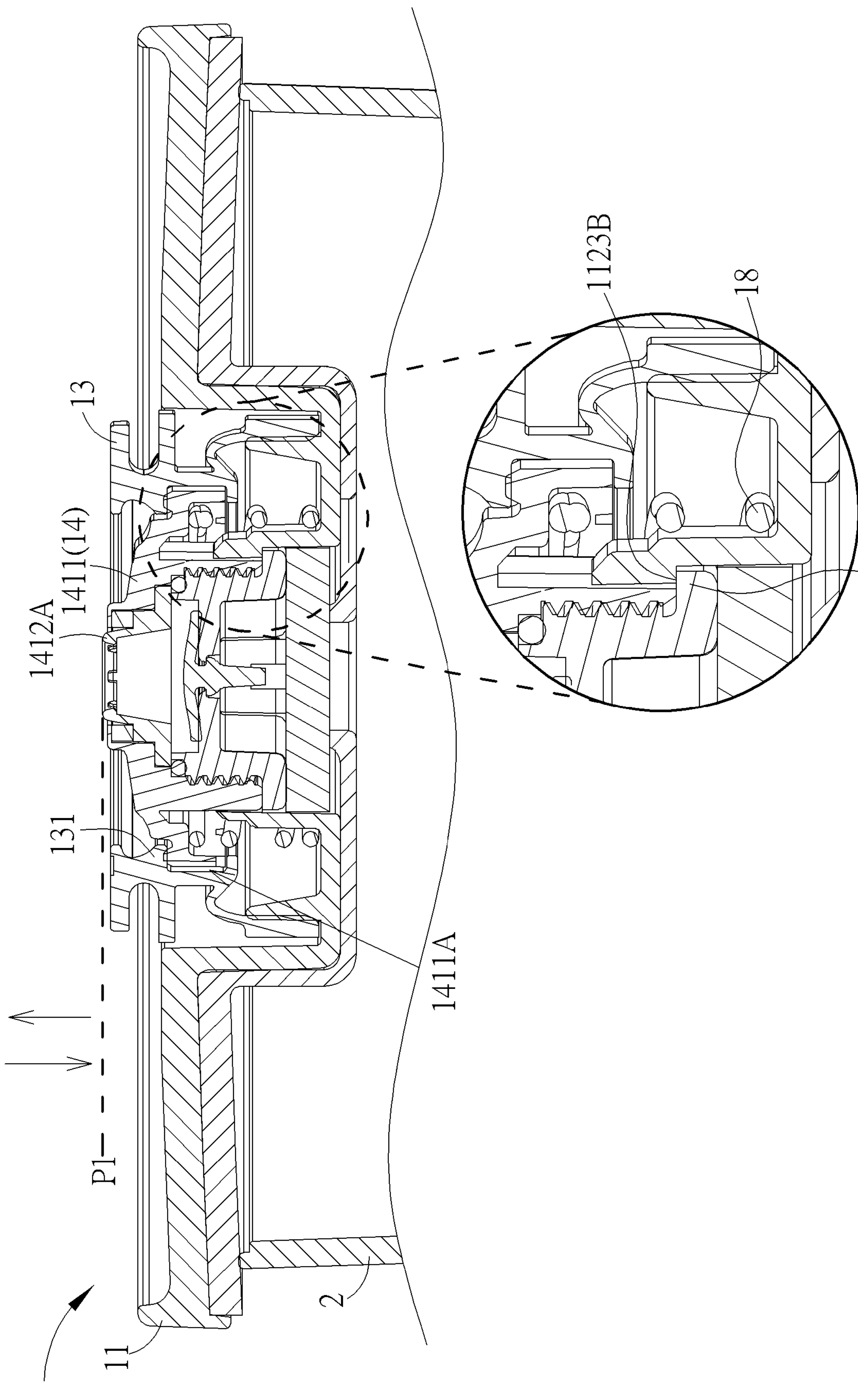


FIG. 7

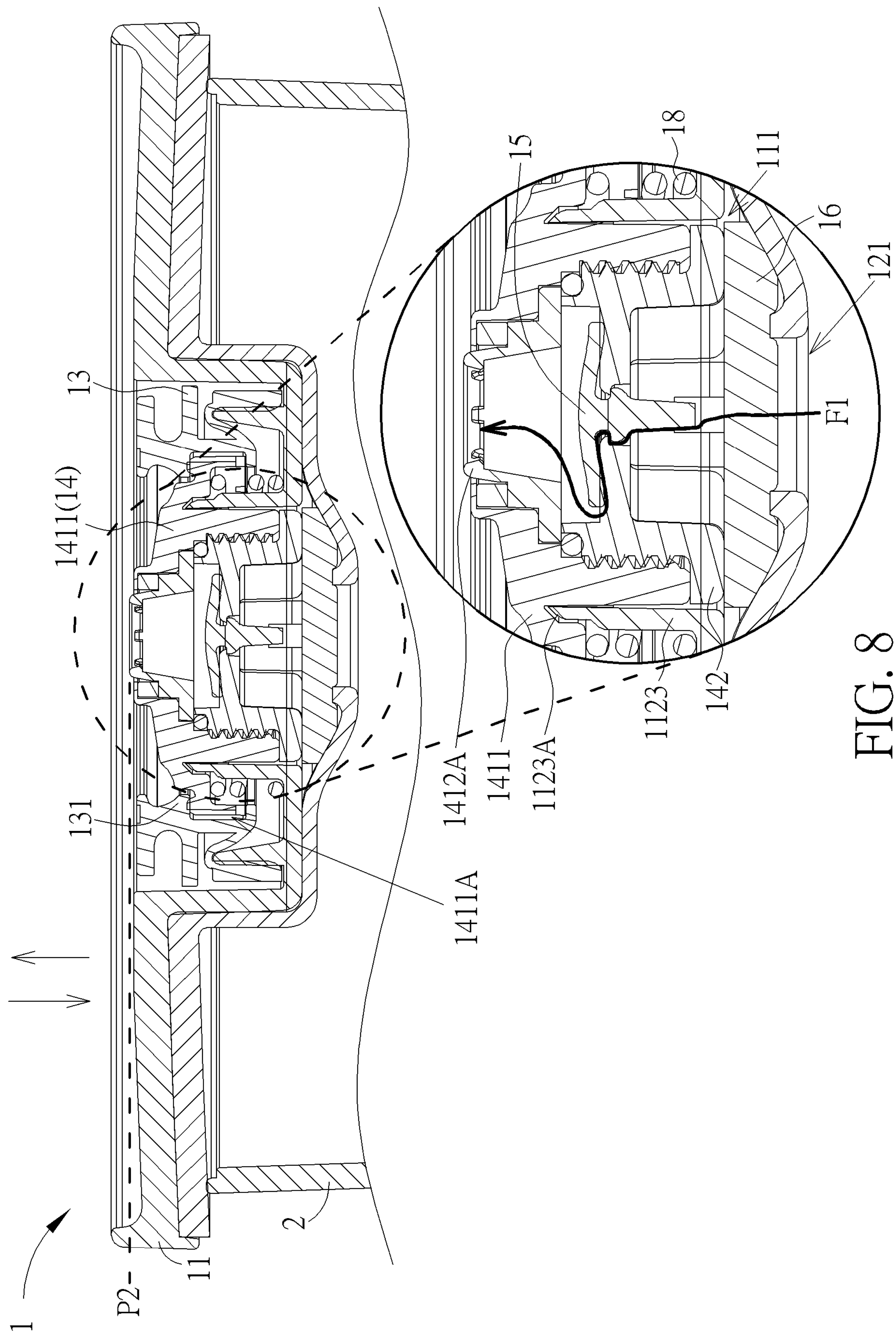

13
1411(14)
131
1411A
15
18
111
16
121
F1
1412A
1411
1123A
1123
142
2
11
P2
1

FIG. 8

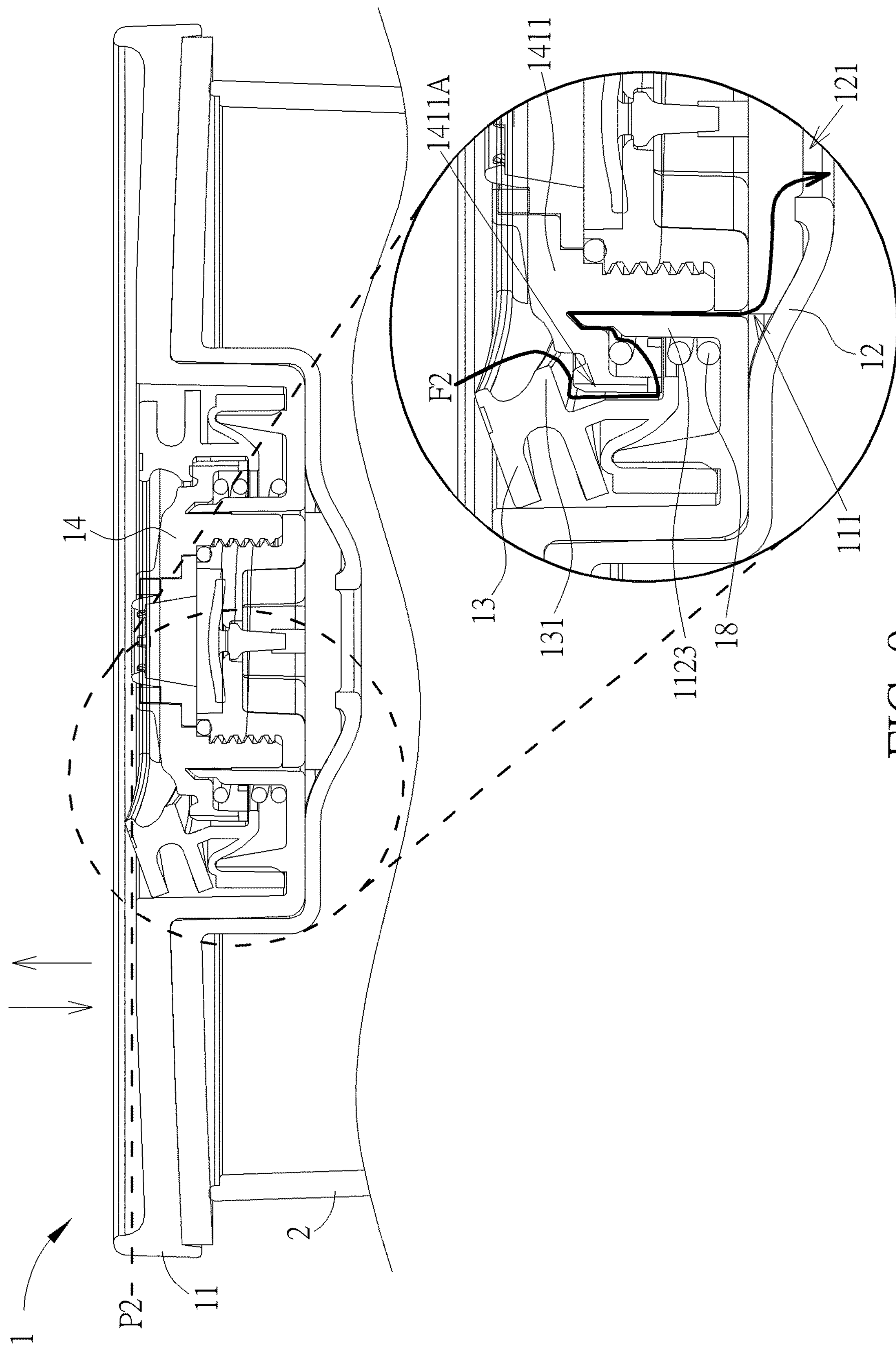


FIG. 9

COVER DEVICE

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to a sealing product, and more specifically, to a cover device adapted for a container.

2. Description of the Prior Art

There are various sealing products available in the market. A sealing cover is one of the sealing products. The sealing cover can be placed on a container, such as a cup, and connected to a vacuum device, such as a vacuum pump, so as to create a vacuum environment or a negative pressure environment inside the container for extending a storage time of objects stored inside the container. However, the conventional sealing cover lacks simple and intuitive use. Therefore, an improvement of the sealing cover is urgently needed.

SUMMARY OF THE INVENTION

It is an objective of the present invention to provide a cover device adapted for a container for solving the aforementioned problems.

In order to achieve the aforementioned objective, the present invention discloses a cover device. The cover device includes a first body, a second body, a resilient component, a seat assembly, a check valve and a filtering component. The second body is removably attached on the first body and resiliently deformable. The resilient component is disposed on the first body. The seat assembly is movable relative to the first body. The seat assembly includes a first seat and a second seat. The first seat is removably engaged with the resilient component and configured to be connected to a vacuum device. The second seat is removably engaged with the first seat. The check valve is disposed on the second seat. The filtering component is disposed between the second seat and the second body.

According to an embodiment of the present invention, the first seat includes an engaging component and a connecting component. The engaging component is removably engaged with the resilient component. The connecting component is removably engaged with the engaging component, and the connecting component includes a connecting structure protruding out of the engaging component and configured to be connected to the vacuum device.

According to an embodiment of the present invention, the cover device further includes a recovering component disposed between the first body and the engaging component and for recovering the engaging component.

According to an embodiment of the present invention, the first seat further includes a magnetic component disposed between the engaging component and the connecting component.

According to an embodiment of the present invention, at least one channel is formed on an outer periphery of the engaging component. The resilient component includes at least one inner rib configured to engage with the outer periphery of the engaging component for blocking the at least one channel, and the resilient component is resiliently deformed to disengage the at least one inner rib from the outer periphery of the engaging component so that the at least one inner rib does not block the at least one channel.

According to an embodiment of the present invention, the first body includes a recess structure. The recess structure includes a base, a first wall and a second wall. A through hole is formed on the base. The first wall is adjacent to an outer periphery of the base and configured to removably engage with the resilient component. The second wall is adjacent to an inner periphery of the base and surrounds the through hole, and the second wall is configured to cooperate with the seat assembly.

According to an embodiment of the present invention, the cover device further includes a recovering component sleeved on the second wall and abutting against the base and the engaging component, and the recovering component is configured to recover the engaging component.

According to an embodiment of the present invention, the seat assembly is movable between a first position and a second position. The second wall includes a first stopping portion and a second stopping portion. The first stopping portion abuts against the engaging component when the seat assembly is located at the second position, and the second stopping portion abuts against the second seat when the seat assembly is located at the first position.

According to an embodiment of the present invention, at least one first guiding structure is arranged on the engaging component, and at least one second guiding structure is arranged on the second wall and configured to cooperate with the at least one first guiding structure for preventing a rotating movement of the engaging component.

According to an embodiment of the present invention, the filtering component is driven by the second seat to at least partially protrude out of a space defined by the second wall through the through hole, so as to resiliently deform the second body.

According to an embodiment of the present invention, the cover device further includes a sealing component disposed in a space defined by the engaging component, the connecting component and the second seat.

According to an embodiment of the present invention, the cover device further includes a recovering component disposed between the first body and the first seat and for recovering the first seat.

According to an embodiment of the present invention, at least one channel is formed on an outer periphery of the first seat. The resilient component includes at least one inner rib configured to engage with the outer periphery of the first seat for blocking the at least one channel, and the resilient component is resiliently deformed to disengage the at least one inner rib from the outer periphery of the first seat so that the at least one inner rib does not block the at least one channel.

According to an embodiment of the present invention, the first body includes a recess structure. The recess structure includes a base, a first wall and a second wall. A through hole is formed on the base. The first wall is adjacent to an outer periphery of the base and configured to removably engage with the resilient component. The second wall is adjacent to an inner periphery of the base and surrounds the through hole, and the second wall is configured to cooperate with the seat assembly.

According to an embodiment of the present invention, the cover device further includes a recovering component sleeved on the second wall and abutting against the base and the first seat, and the recovering component is configured to recover the first seat.

According to an embodiment of the present invention, the seat assembly is movable between a first position and a second position. The second wall includes a first stopping portion and a second stopping portion. The first stopping portion abuts against the first seat when the seat assembly is located at the second position, and the second stopping portion abuts against the second seat when the seat assembly is located at the first position.

According to an embodiment of the present invention, at least one first guiding structure is arranged on the first seat, and at least one second guiding structure is arranged on the second wall and configured to cooperate with the at least one first guiding structure for preventing a rotating movement of the first seat.

According to an embodiment of the present invention, the filtering component is driven by the second seat to at least partially protrude out of the first body, so as to resiliently deform the second body.

According to an embodiment of the present invention, the seat assembly is movable between a first position and a second position. The first body includes a first stopping portion and a second stopping portion. The first stopping portion abuts against the first seat when the seat assembly is located at the second position. The second stopping portion abuts against the second seat when the seat assembly is located at the first position. At least one first guiding structure is arranged on the first seat, and at least one second guiding structure is arranged on the first body and configured to cooperate with the at least one first guiding structure for preventing a rotating movement of the first seat.

According to an embodiment of the present invention, the cover device further includes a sealing component disposed in a space defined by the first seat and the second seat.

In summary, the present invention enables the seat assembly to move from a higher position, e.g., the first position, to a lower position, e.g., the second position, when a pressure of an interior of the container gradually decreases. Besides, the present invention further enables the seat assembly to move from the lower position, e.g., the second position, to the higher position, e.g., the first position, when the pressure of the interior of the container gradually increases. Therefore, the pressure of the interior of the container can be estimated by observing a position of the seat assembly easily. Besides, an air passage can be formed for allowing ambient air to flow into the interior of the container when the resilient component is resiliently deformed to drive at least a portion of the resilient component, e.g., the inner rib of the resilient component, and at least a portion of the seat assembly, e.g., the outer periphery of the engaging component of the first seat of the seat assembly, to disengage from each other. Therefore, the present invention has intuitive use.

These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a diagram of a cover device attached on a container according to an embodiment of the present invention.

FIG. **2** is a diagram of the cover device at another view according to the embodiment of the present invention.

FIG. **3** is a sectional diagram of the cover device according to the embodiment of the present invention.

FIG. **4** and FIG. **5** are partial exploded diagrams of the cover device at different views according to the embodiment of the present invention.

FIG. **6** is a sectional exploded diagram of the cover device according to the embodiment of the present invention.

FIG. **7** is a diagram of the cover device in a first used state according to the embodiment of the present invention.

FIG. **8** is a diagram of the cover device in a second used state according to the embodiment of the present invention.

FIG. **9** is a diagram of the cover device in a third used state according to the embodiment of the present invention.

DETAILED DESCRIPTION

In the following detailed description of the preferred embodiments, reference is made to the accompanying drawings which form a part hereof, and in which is shown by way of illustration specific embodiments in which the invention may be practiced. In this regard, directional terminology, such as "top", "bottom", "left", "right", "front", "back", etc., is used with reference to the orientation of the Figure(s) being described. The components of the present invention can be positioned in a number of different orientations. As such, the directional terminology is used for purposes of illustration and is in no way limiting. Accordingly, the drawings and descriptions will be regarded as illustrative in nature and not as restrictive. Also, if not specified, the term "connect" is intended to mean either an indirect or direct mechanical connection. Thus, if a first device is connected to a second device, that connection may be through a direct mechanical connection, or through an indirect mechanical connection via other devices and connections.

Please refer to FIG. **1** to FIG. **3**. FIG. **1** is a diagram of a cover device **1** attached on a container **2** according to an embodiment of the present invention. FIG. **2** is a diagram of the cover device **1** at another view according to the embodiment of the present invention. FIG. **3** is a sectional diagram of the cover device **1** according to the embodiment of the present invention. As shown in FIG. **1** to FIG. **3**, the cover device **1** includes a first body **11**, a second body **12**, a resilient component **13**, a seat assembly **14**, a check valve **15** and a filtering component **16**. The second body **12** is removably attached on the first body **11** and resiliently deformable. The resilient component **13** is disposed on the first body **11**. The seat assembly **14** is movable relative to the first body **11**. The seat assembly **14** includes a first seat **141** and a second seat **142**. The first seat **141** is removably engaged with the resilient component **13** and configured to be connected to a vacuum device, e.g., a vacuum pump, which is not shown in the figures and is configured to draw air out of an interior of the container **2**. The second seat **142** is removably engaged with the first seat **141**. The check valve **15** is disposed on the second seat **142**. The filtering component **16** is disposed between the second seat **142** and the second body **12**.

In this embodiment, the second body **12** and the resilient component **13** can be made of silicone rubber. The check valve **15** can be an umbrella valve. The filtering component **16** can be a foam component. However, the present invention is not limited to this embodiment.

Specifically, as shown in FIGS. **2** and **3**, at least one communication hole **121** is formed on the second body **12**, and at least one through hole **111** is formed on the first body **11**. The check valve **15** is for allowing the air to flow from the interior of the container **2** through the at least one communication hole **121** and the at least one through hole **111** and preventing ambient air from flowing into the interior of the container **2**. The filtering component **16** is disposed between the check valve **15** and the at least one through hole **111** for filtering particles in the air flowing from the interior of the container **2** through the at least one communication hole **121** and the at least one through hole **111**, so as to prevent damage of the check valve **15** caused by the particles.

Please refer to FIG. **3** to FIG. **6**. FIG. **4** and FIG. **5** are partial exploded diagrams of the cover device **1** at different views according to the embodiment of the present invention. FIG. **6** is a sectional exploded diagram of the cover device **1** according to the embodiment of the present invention. As shown in FIG. **3** to FIG. **6**, the first seat **141** includes an engaging component **1411** and a connecting component **1412**. The engaging component **1411** is removably engaged with the resilient component **13**. The connecting component **1412** is removably engaged with the engaging component **1411**, and the connecting component **1412** includes a connecting structure **1412**A protruding out of the engaging component **1411** and configured to be connected to the vacuum device.

As shown in FIG. **3** to FIG. **6**, in this embodiment, the first seat **141** further includes a magnetic component **1413** disposed between the engaging component **1411** and the connecting component **1412** to magnetically attract the vacuum device for facilitating a connection of the connecting structure **1412**A and the vacuum device.

However, the present invention is not limited to this embodiment. For example, in another embodiment, the magnetic component can be omitted, and the engaging component and the connecting component can be integrally formed with each other. Alternatively, in another embodiment, the engaging component and the connecting component can be integrally formed with each other, and the magnetic component can be embedded inside an integral structure formed by the engaging component and the connecting component.

Besides, as shown in FIG. **3** to FIG. **5**, in this embodiment, the second seat **142** can be rotatably engaged with the first seat **141** and rotatably disengaged from the first seat **141** by a cooperation of an outer thread structure on the second seat **142** and an inner thread structure on the engaging component **1411**.

However, the present invention is not limited to this embodiment. For example, in another embodiment, the second seat can be removably engaged with the first seat by a cooperation of an engaging hook and an engaging slot formed on one of the second seat and the first seat and the other of the second seat and the first seat respectively, or by a cooperation of two magnetic components embedded inside the second seat and the first seat respectively.

As shown in FIG. **3**, in this embodiment, the cover device **1** further includes a sealing component **17** disposed in a space defined by the first seat **141** and the second seat **142**. Specifically, the sealing component **17** can be disposed in a space defined by the engaging component **1411**, the connecting component **1412** and the second seat **142** for preventing air leakage through a gap between the engaging component **1411** and the connecting component **1412** and/or a gap between the engaging component **1411** and the second seat **142**.

However, the present invention is not limited to this embodiment. For example, in another embodiment, the sealing component can be disposed in a space only defined by the engaging component and the second seat. Alternatively, in another embodiment, the sealing component can be omitted.

As shown in FIG. **3**, FIG. **4** and FIG. **6**, the first body **11** includes a recess structure **112**. The recess structure **112** includes a base **1121**, a first wall **1122** and a second wall **1123**. The at least one through hole **111** is formed on the base **1121**. The first wall **1122** is adjacent to an outer periphery of the base **1121** and configured to removably engage with the resilient component **13**. The second wall **1123** is adjacent to an inner periphery of the base **1121** and surrounds the at least one through hole **111**, and the second wall **1123** is configured to cooperate with the seat assembly **14**.

In addition, as shown in FIG. **3**, FIG. **4** and FIG. **6**, in this embodiment, the cover device **1** further includes a recovering component **18** disposed between the first body **11** and the first seat **141** and for recovering the first seat **141**. Specifically, the recovering component **18** can be sleeved on the second wall **1123** and abut against the base **1121** and the engaging component **1411**.

However, the present invention is not limited to this embodiment. For example, in another embodiment, the first seat can be driven to recover by a resilient force provided by the resilient component, and the recovering component can be omitted.

As shown in FIG. **6**, in this embodiment, the seat assembly **14** is only allowed to move within a limited range during a normal operation. Specifically, the second wall **1123** includes a first stopping portion **1123**A and a second stopping portion **1123**B for respectively abutting against the engaging component **1411** and the second seat **142** to restrain a traveling distance of the seat assembly **14**.

However, the present invention is not limited to this embodiment. For example, in another embodiment, the first stopping portion and the second stopping portion can be omitted.

Furthermore, as shown in FIG. **4** to FIG. **6**, in order to facilitate an engagement of the first seat **141** and the second seat **142** and ensure a stable movement of the seat assembly **14**, at least one first guiding structure **1411**B is arranged on the engaging component **1411**, and at least one second guiding structure **1123**C is arranged on the second wall **1123** and configured to cooperate with the at least one first guiding structure **1411**B for preventing a rotating movement of the engaging component **1411** when the seat assembly **14** moves relative to the first body **11** and when the second seat **142** is engaged with the engaging component **1411**.

As shown in FIG. **3** to FIG. **6**, in order to achieve easy pressure relief, at least one channel **1411**A is formed on an outer periphery of the engaging component **1411**. The resilient component **13** includes at least one inner rib **131** configured to engage with the outer periphery of the engaging component **1411** for blocking the at least one channel **1411**A, and the resilient component **13** is resiliently deformed to disengage the at least one inner rib **131** from the outer periphery of the engaging component **1411** so that the at least one inner rib **131** does not block the at least one channel **1411**A to allow pressure relief.

Detailed description for the operational principle of the cover device **1** is provided as follows. FIG. **7** is a diagram of the cover device **1** in a first used state according to the embodiment of the present invention. FIG. **8** is a diagram of the cover device **1** in a second used state according to the embodiment of the present invention. FIG. **9** is a diagram of the cover device **1** in a third used state according to the embodiment of the present invention. Before the vacuum device is connected to the connecting structure **1412**A, or when the vacuum device is not activated for drawing the air out of the interior of the container **2**, a pressure of the interior of the container **2** is substantially equal to an ambient pressure. Therefore, at this moment, the cover device **1** is in the first used state, i.e., the seat assembly **14** is driven to the first position P**1** by the recovering component **18** and/or the resilient component **13**, and the second stopping portion **1123**B abuts against the second seat **142** for positioning the seat assembly **14** at the first position P**1**. After the vacuum device is connected to the connecting structure **1412**A and activated for drawing the air out of the interior of the container **2** through a first path F**1** as shown in FIG. **8**, i.e., through the at least one communication hole **121**, the at least one through hole **111**, at least one gap between the check valve **15** and the second seat **142**, and the connecting structure **1412**A, the pressure of the interior of the container **2** gradually decreases, such that a difference between the pressure of the interior of the container **2** and the ambient pressure causes the seat assembly **14** to move from the first position P**1** as shown in FIG. **7** to a second position P**2** as shown in FIG. **8** and to resiliently deform the recovering component **18** and the resilient component **13**. When the seat assembly **14** is located the second position P**2** as shown in FIG. **8**, the first stopping portion **1123**A abuts against the engaging component **1411** for positioning the seat assembly **14** at the second position P**2**, and the filtering component **16** is resiliently deformed and is driven by the second seat **142** to at least partially protrude out of a space defined by the second wall **1123** through the through hole **111**, so as to resiliently deform the second body **12**. Therefore, the pressure of the interior of the container **2** can be estimated by observing a position of the seat assembly **14** easily. During the aforementioned process, the at least one inner rib **131** keeps engaging with the outer periphery of the engaging component **1411** to block the at least one channel **1411**, which facilitates drawing the air out of the interior of the container **2**.

As shown in FIG. **9**, when it is desired to detach the cover device **1** from the container **2**, the resilient component **13** can be resiliently deformed, e.g., by an external force, to disengage the at least one inner rib **131** from the outer periphery of the engaging component **1411** so that the at least one inner rib **131** does not block the at least one channel **1411**A, so as to allow the ambient air to flow into the interior of the container **2** through a second path F**2**, i.e., through the at least one channel **1411**A, at least one gap between the seat assembly **14** and the second wall **1123**, the at least one through hole **111**, and the at least one communication hole **121**, for pressure relief. During the aforementioned pressure relief process, the pressure of the interior of the container **2** gradually increases until the pressure of the interior of the container **2** is equal to the ambient pressure, i.e., the difference between the pressure of the interior of the container **2** and the ambient pressure is reduced to zero substantially, such that the seat assembly **14** is driven to move from the second position P**2** as shown in FIG. **8** back to the first position P**1** as shown in FIG. **7** by the recovering component **18** and/or the resilient component **13** and positioned at the first position P**1** by an abutment of the second stopping portion **1123**B and the second seat **142**. After the seat assembly **14** moves to the first position P**1** as shown in FIG. **7**, the cover device **1** can be detached from the container **2** easily.

In contrast to the prior art, the present invention enables the seat assembly to move from a higher position, e.g., the first position, to a lower position, e.g., the second position, when the pressure of the interior of the container gradually decreases. Besides, the present invention further enables the seat assembly to move from the lower position, e.g., the second position, to the higher position, e.g., the first position, when the pressure of the interior of the container gradually increases. Therefore, the pressure of the interior of the container can be estimated by observing the position of the seat assembly easily. Besides, an air passage can be formed for allowing ambient air to flow into the interior of the container when the resilient component is resiliently deformed to drive at least a portion of the resilient component, e.g., the inner rib of the resilient component, and at least a portion of the seat assembly, e.g., the outer periphery of the engaging component of the first seat of the seat assembly, to disengage from each other. Therefore, the present invention has simple and intuitive use.

Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

What is claimed is:

1. A cover device comprising:
a first body;
a second body removably attached on the first body and resiliently deformable;
a resilient component disposed on the first body;
a seat assembly movable relative to the first body, the seat assembly comprising:
a first seat removably engaged with the resilient component and configured to be connected to a vacuum device; and
a second seat removably engaged with the first seat;
a check valve disposed on the second seat; and
a filtering component disposed between the second seat and the second body.

2. The cover device of claim **1**, wherein the first seat comprises an engaging component and a connecting component, the engaging component is removably engaged with the resilient component, the connecting component is removably engaged with the engaging component, and the connecting component comprises a connecting structure protruding out of the engaging component and configured to be connected to the vacuum device.

3. The cover device of claim **2**, further comprising a recovering component disposed between the first body and the engaging component and for recovering the engaging component.

4. The cover device of claim **2**, wherein the first seat further comprises a magnetic component disposed between the engaging component and the connecting component.

5. The cover device of claim **2**, wherein at least one channel is formed on an outer periphery of the engaging component, the resilient component comprises at least one inner rib configured to engage with the outer periphery of the engaging component for blocking the at least one channel, and the resilient component is resiliently deformed to disengage the at least one inner rib from the outer periphery of the engaging component so that the at least one inner rib does not block the at least one channel.

6. The cover device of claim **2**, wherein the first body comprises a recess structure, the recess structure comprises a base, a first wall and a second wall, a through hole is formed on the base, the first wall is adjacent to an outer periphery of the base and configured to removably engage with the resilient component, the second wall is adjacent to an inner periphery of the base and surrounds the through hole, and the second wall is configured to cooperate with the seat assembly.

7. The cover device of claim **6**, further comprising a recovering component sleeved on the second wall and abutting against the base and the engaging component, and the recovering component being configured to recover the engaging component.

8. The cover device of claim **6**, wherein the seat assembly is movable between a first position and a second position, the second wall comprises a first stopping portion and a second stopping portion, the first stopping portion abuts against the engaging component when the seat assembly is located at the second position, and the second stopping portion abuts against the second seat when the seat assembly is located at the first position.

9. The cover device of claim **6**, wherein at least one first guiding structure is arranged on the engaging component, and at least one second guiding structure is arranged on the second wall and configured to cooperate with the at least one first guiding structure for preventing a rotating movement of the engaging component.

10. The cover device of claim **6**, wherein the filtering component is driven by the second seat to at least partially protrude out of a space defined by the second wall through the through hole, so as to resiliently deform the second body.

11. The cover device of claim **2**, further comprising a sealing component disposed in a space defined by the engaging component, the connecting component and the second seat.

12. The cover device of claim **1**, further comprising a recovering component disposed between the first body and the first seat and for recovering the first seat.

13. The cover device of claim **1**, wherein at least one channel is formed on an outer periphery of the first seat, the resilient component comprises at least one inner rib configured to engage with the outer periphery of the first seat for blocking the at least one channel, and the resilient component is resiliently deformed to disengage the at least one inner rib from the outer periphery of the first seat so that the at least one inner rib does not block the at least one channel.

14. The cover device of claim **1**, wherein the first body comprises a recess structure, the recess structure comprises a base, a first wall and a second wall, a through hole is formed on the base, the first wall is adjacent to an outer periphery of the base and configured to removably engage with the resilient component, the second wall is adjacent to an inner periphery of the base and surrounds the through hole, and the second wall is configured to cooperate with the seat assembly.

15. The cover device of claim **14**, further comprising a recovering component sleeved on the second wall and abutting against the base and the first seat, and the recovering component being configured to recover the first seat.

16. The cover device of claim **14**, wherein the seat assembly is movable between a first position and a second position, the second wall comprises a first stopping portion and a second stopping portion, the first stopping portion abuts against the first seat when the seat assembly is located at the second position, and the second stopping portion abuts against the second seat when the seat assembly is located at the first position.

17. The cover device of claim **14**, wherein at least one first guiding structure is arranged on the first seat, and at least one second guiding structure is arranged on the second wall and configured to cooperate with the at least one first guiding structure for preventing a rotating movement of the first seat.

18. The cover device of claim **1**, wherein the filtering component is driven by the second seat to at least partially protrude out of the first body, so as to resiliently deform the second body.

19. The cover device of claim **1**, wherein the seat assembly is movable between a first position and a second position, the first body comprises a first stopping portion and a second stopping portion, the first stopping portion abuts against the first seat when the seat assembly is located at the second position, the second stopping portion abuts against the second seat when the seat assembly is located at the first position, at least one first guiding structure is arranged on the first seat, and at least one second guiding structure is arranged on the first body and configured to cooperate with the at least one first guiding structure for preventing a rotating movement of the first seat.

20. The cover device of claim **1**, further comprising a sealing component disposed in a space defined by the first seat and the second seat.

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